



Poverty, Toxic Stress, and Education in Children Born Preterm

Michelle M. Kelly ▼ Kati Li

Background: Preterm birth, defined as birth before the completion of 37 weeks of gestation, is a multifactorial global epidemic with psychosocial, economic, and physical ramifications affecting the child, family, and community at large. Toxic stress—the results of exposure to adverse childhood experiences—results in changes to brain structure and function that negatively affects future health.

Objectives: The aim of this study was to apply the eco-bio-developmental (EBD) model of poverty and preterm birth to the cross-sectional data of the 2016 National Survey of Children's Health to evaluate the associations between poverty, toxic stress, and prematurity on neurodevelopmental and educational outcomes.

Methods: A subset of data representing children ages 6–11 years old ($n = 15,010$) from the 2016 National Survey of Children's Health was used for multivariate analysis of demographic variables and neurodevelopmental and educational outcome variables. Pearson's chi-square, logistic regression, and interaction effects explored the relationships between prematurity, toxic stress, and poverty.

Results: Children in this sample born preterm had a higher incidence of toxic stress, poverty, developmental delay, learning disability, intellectual disability, speech/language disorders, attention-deficit disorder (ADD)/attention-deficit/hyperactivity disorder (ADHD), autism, and special education/early intervention plans. The combination of poverty, toxic stress, and preterm birth significantly increased the risk of these conditions. After accounting for gender, insurance coverage, race, and parental education, children in the sample born preterm were more likely to experience developmental delay, intellectual disability, speech/language disorder, learning disability, and ADD/ADHD. Toxic stress increased the incidence of ADD/ADHD and autism in both the preterm and full-term samples.

Discussion: The negative effect of poverty and toxic stress on children born preterm, as depicted by the eco-bio-developmental model, is supported by this analysis. Healthcare providers are encouraged to address the tripartite vulnerability resulting from prematurity, poverty, and toxic stress.

Key Words: poverty • prematurity • toxic stress

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Preterm birth, defined as birth before the completion of 37 weeks of gestation, is a multifactorial global epidemic with psychosocial, economic, and physical ramifications affecting the child, family, and community at large (Dempsey et al., 2015; Harrison & Goldberg, 2016). Studies of long-term follow-up suggest that individuals born preterm are at risk for decreased academic achievement, lower adult wages, and decreased adult wealth and that these results extend to those born 32–26 weeks gestational age (Basten, Jaekel, Johnson, Gilmore, & Wolke, 2015; Johnson & Marlow, 2017). Early physiological insults in the perinatal period negatively affect the physical, cognitive, and emotional development over the individual's life span (Litt & McCormick, 2016). Analysis

of over 4.1 million singleton births from five countries with very high human development index suggests that two thirds of all preterm births have no biological explanation (Ferrero et al., 2016). Ferrero and colleagues selected countries that scored very high on the human development index, a geometric mean of key dimensions of human development: A long and healthy life, being knowledgeable, and having a decent standard of living advocated for use by the United Nations, as more robust than measures of economic growth alone (<http://hdr.undp.org/en/content/human-development-index-hdi>). The inability to predict the biological etiology of preterm birth ensures that children will continue to be born preterm despite ongoing medical advances in survival and treatment. In the United States, the preterm birth rate hovers just below 10%, with consecutive increases between 2014 and 2017 attributed to the increase in children born between 34 and 36 weeks of gestation (Hamilton, Martin, Osterman, Driscoll, & Rossen, 2018).

Adverse childhood experiences (ACEs) are individual characteristics relating to the home and family environment.

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ACEs include abuse (emotional, physical, or psychological), emotional and physical neglect, family dysfunctions including alcohol and drug abuse, divorce or loss of a biological parent, depression or mental illness in the home, violent treatment of the mother, and caregiver incarceration (Biglan, Van Ryzin, & Hawkins, 2017). The original ACE research (Felitti et al., 1998) and subsequent published research (e.g., Biglan et al., 2017) showed that exposure to ACEs resulted in greater likelihood of premature death due to physical illness. This risk is separate from the risk associated with poverty, as escaping poverty did not attenuate the risk from ACE (Biglan et al., 2017).

It is thought that ACEs result in frequent and prolonged activation of a body's stress response system and can induce a toxic stress response (Shonkoff, Garner, Committee on Psychosocial Aspects of Child and Family Health, Committee on Early Childhood, Adoption, and Dependent Care, & Section on Developmental and Behavioral Pediatrics, 2012). Children who experience ACEs without the buffering effect of supportive adult relationships are particularly vulnerable to the disruption of brain circuitry and organ and metabolic systems (Shonkoff et al., 2012). Toxic stress is believed to disrupt brain architecture, alter organ systems, and negatively affect stress regulation systems and the capacity to adapt to future adversity (Shonkoff, Boyce, & McEwen, 2009; Shonkoff et al., 2012). The American Academy of Pediatrics released a technical report in 2012 focusing on the persistent health effects of ACEs and the resulting toxic stress, calling for coordinated efforts to combat the social conditions that result in exposure to ACE (Shonkoff et al., 2012).

The eco-bio-developmental (EBD) framework highlights the implications of ACEs and the resulting toxic stress on the developing brain and their effect on future behavioral, educational, economic, and health outcomes (Garner, Shonkoff, Committee on Psychosocial Aspects of Child and Family Health, Committee on Early Childhood, Adoption, and Dependent Care, & Section on Developmental and Behavioral Pediatrics, 2012). Incorporation of this framework into current healthcare practice requires the recognition that early stress experiences may undermine the adaptive capacity needed to deal with later challenges, that maladaptive coping patterns may be behavioral and psychological responses that emerge in early childhood, and that prevention of adverse consequences starts with stable responsive relationships that facilitate a restoration of stress responses (Garner et al., 2012). The EBD framework was adapted by the American Academy of Pediatrics from the science-based framework created by the Center on the Developing Child and aimed at advancing policies and programs to improve child health (Shonkoff et al., 2009).

APPLYING EBD THEORETICAL FRAMEWORK TO CHILDREN BORN PRETERM

Recently, researchers have explored the relationship between a woman's exposure to ACEs and the incidence of preterm

delivery. Christiaens, Hegadoren, and Olson (2015) conducted a case-control study of women with preterm (prior to 37 weeks gestation) singleton delivery without premature rupture of membranes ($n = 75$) and a control group of women with singleton delivery at term without prior history of preterm birth ($n = 148$). Exposure to two or more ACEs was associated with preterm birth regardless of demographic and clinical risk factors (maternal age, smoking status, educational status, and history of miscarriage; Christiaens et al., 2015). After confounding variables were controlled, each additional ACE increased the risk of preterm birth by 18%. Lifetime abuse (childhood and adult) also increased the risk of preterm birth.

Similarly, Smith, Gotman, and Yonkers (2016) conducted a prospective study of 2,302 pregnant women and analyzed the association of ACE and pregnancy outcomes. After controlling for mediators, each ACE accounted for a 13.79-g decrease in birth weight, and a 0.041-week decrease in gestational age (Smith et al., 2016). Following adjustment of other mediators, maternal smoking accounted for significant decreases in birth weight and gestational age and was identified more often in pregnant women who experienced ACE (Smith et al., 2016).

Other researchers have used a bioecological framework to address the academic outcomes of children born preterm. Litt and McCormick (2016) analyzed the interplay of preterm birth and special healthcare needs at school age on academic success, through a bioecological framework. Experiencing a special healthcare needs at school age was shown to have a negative effect on math and reading scores at 8 and 18 years of age; this effect was increased with earlier gestational ages (Litt & McCormick, 2016). Math and reading scores were found to decrease between the 8 and 18 years of age assessments. The researchers concluded that middle childhood represents a critical period of literacy and numeracy skill attainment required for higher level tasks. In a United States-based sample of children born preterm stratified by neonatal morbidity, Sullivan, Miller, and Msall (2012) determined that social capital, not neonatal morbidity, had the greatest effect on cognition, learning, and academic outcomes at the age of 17 years.

Furthering the application of the EBD framework, Brumberg and Shah (2015) applied the framework to children born preterm. Using the resulting EBD model of poverty and preterm birth, the researchers hypothesized that poverty and toxic stress occurring prior to conception or during pregnancy predispose the mother to premature deliver; subsequently, the child born preterm who also experiences poverty and toxic stress suffers from alterations in brain function that leads to impaired neurological outcomes. As the child ages, these changes lead to decreased educational attainment and adult poverty, which restarts the cascade in the preterm child who is now an adult (see SDC Figure 1, <http://links.lww.com/NRES/A316>).

Extensive research supports the detrimental nature of poverty and toxic stress on the health and neurodevelopment of children. However, existing literature fails to adequately

explore the effect of poverty and toxic stress on the neurodevelopmental and educational outcomes of a specific group of children: those born preterm. In the current research, we applied the EBD model of poverty and preterm birth to the cross-sectional data of the 2016 National Survey of Children's Health (NSCH) to evaluate the associations between poverty, toxic stress, and prematurity on neurodevelopmental and educational outcomes.

METHODS

Data

The 2016 NSCH data set used in this study was compiled by the U.S. Census Bureau. The survey was funded by the U.S. Department of Health and Human Services Health Resources and Services Administration's Maternal and Child Health Bureau. The NSCH is designed to produce national and state-level data on the physical and emotional health of American children 0–17 years old (Child and Adolescent Health Measurement Initiative [CAHMI] & Data Resource Center for Child and Adolescent Health [DRCCAH], 2018). Information was collected on factors related to the health and well-being of children, including access to and utilization of healthcare, receipt of care in a medical home, family interactions, parental health, school and after-school experiences, and neighborhood characteristics (CAHMI & DRCCAH, 2018).

The 2016 NSCH used a two-phase, self-administered data collection design with Internet-based web and mailed paper data collection instruments. A parent or guardian with knowledge of the health and healthcare of the sampled child in the household completed an initial screener with the age and gender of all children in the household; one child from each household was then randomly selected to be the subject of the main topical questionnaire (CAHMI & DRCCAH, 2018). A total of 139,923 households in the 50 states and the District of Columbia were screened for age-eligible children. Of these households, 68,961 reported age-eligible children living or staying in the household (CAHMI & DRCCAH, 2018). From this sample of households with age-eligible children, 50,212 detailed topical questionnaires were completed. The weighted overall response rate was 40.7%. The NSCH 2016 used age-specific questionnaires for three age subsets: 0–5 years, 6–11 years, and 12–17 years. In the analysis reported here, we used the subset of children 6–11 years old ($n = 15,010$). This subset was selected to reflect the experiences of children during elementary school. The Villanova University Institutional Review Board evaluated the research plan and deemed the study not subject to federal regulations under HHS 45 CFR 46.

Measures

Predictor variables included poverty, toxic stress, and prematurity. Poverty was defined as living in a household with an

income of less than 100% of the federal poverty level. The federal poverty level numbers are based on the federal poverty guidelines issued annually by the Department of Health and Human Services. A child was identified as having experienced toxic stress if his or her parent reported that the child had experienced one or more ACE from the following list: It was “somewhat often or very often hard to get by on family income”; “parent or guardian divorced or separated”; “parent or guardian died”; “parent or guardian served time in jail”; “saw or heard parents or adults slap, hit, kick, punch one another in the home”; “victim/witness of neighborhood violence”; “lived with anyone who was mentally ill, suicidal, or severely depressed”; “lived with anyone who had a problem with alcohol or drug”; and “treated or judged unfairly because of his or her race or ethnic group” (CAHMI & DRCCAH, 2018). Prematurity was defined as a positive response to the question, “Was your child born prematurely, that is, more than 3 weeks before his or her due date?”

Multivariate analyses included the following demographic variables as controls: gender of the child (male, female), insurance coverage status (covered, not covered), race of the child (White alone, Black or African American alone, other), and highest level of education among reported adults primarily involved in the child's care (less than high school, high school or general equivalency diploma [GED], some college or technical school, college degree or higher).

Neurodevelopmental outcome variables used included developmental delay, intellectual disability, speech or other language disorder, learning disability, autism or autism spectrum disorder (ASD), and attention-deficit disorder (ADD)/attention-deficit/hyperactivity disorder (ADHD). Parents were asked two questions relative to the neurodevelopmental conditions: “Has a doctor or other healthcare provider *ever* told you that this child has [condition]?” and “Does this child currently have the condition?” Children were grouped into two categories, having the condition or not having the condition (CAHMI & DRCCAH, 2018).

Educational outcome variables, including repeating a grade and receiving special education/early intervention plan, were considered in this study. A child was counted as having repeated a grade if his or her parent said “yes” to the question, “Since starting kindergarten, has this child repeated any grades?” If the parent said “yes” to the question, “Is this child currently receiving services under a special education or early intervention plan?” the child was counted as having received special education or an early intervention plan (CAHMI & DRCCAH, 2018). An early intervention plan included individualized education plans, 504 Plans, speech language therapy, vision and hearing services, psychological services, health services, social work services, family counseling and support, transportation, or other services needed to support the child's educational performance (CAHMI & DRCCAH, 2018).

Statistical Analyses

Pearson's chi-square test was used to examine associations between prematurity, toxic stress, and neurodevelopmental and educational outcomes. Logistic regression modeling was used to determine the effect of prematurity and toxic stress on neurodevelopmental and educational outcomes. Potential effects of confounding demographic variables were considered (gender, insurance coverage, race, and education of adult caregiver). Odds ratios and 95% confidence intervals (CIs) were computed. To assess whether toxic stress moderates the relationship between prematurity and neurodevelopmental or educational outcomes, interaction effects between prematurity and toxic stress were examined. Statistical analysis was performed with IBM SPSS Statistics for Windows, Version 24.0 (IBM Corp., released 2016, Armonk, NY). All analyses were weighted to account for NSCH's complex survey design and were performed using the SPSS complex sampling design module.

RESULTS

In the 2016 NSCH, there were 15,010 children aged 6–11 years old. Statistical analyses were conducted on the subset of

children for whom full-term/prematurity status was indicated ($n = 14,796$). Table 1 shows the sample characteristics (weighted % and standard error) for the analytic sample as well as the full-term ($n = 13,088$) and preterm ($n = 1,708$) subsamples. These characteristics are included to show that the preterm birth rate of 11.9% seen in the NSCH 2016 data set is consistent with the preterm birth rate reported in the *National Vital Statistics Reports* for those birth years (Martin, Hamilton, Ventura, Osterman, & Matthews, 2013). Of the total sample, 47.4% of the 6- to 11-year-old children experienced toxic stress; 53.9% of the preterm sample experienced toxic stress ($p < .05$), and 46.5% of the full-term sample experienced toxic stress ($p < .05$). Of the total sample, 22.7% of children live in poverty, 27.8% of preterm children live in poverty ($p < .05$), and 22.1% of the full-term children live in poverty. Supplemental analysis revealed that poverty and toxic stress were positively related to one another; that is, children experiencing poverty were 3.0 times more likely to have experienced toxic stress. Parental education attainment, gender, race, and insurance status were consistent between those born preterm and those born full term.

TABLE 1. Characteristics of 6- to 11-Year-Old Sample ($N = 15,010$)

Characteristic	All children % (SE) ($N = 14,796$) ^a	Preterm % (SE) ($n = 1,708$, 11.9%)	Full term % (SE) ($n = 13,088$, 88.1%)
Toxic stress (≥ 1 ACE)	47.4 (1.0)	53.9 (2.7)	46.5 (1.0)*
Poverty	22.7 (0.9)	27.8 (2.9)	22.1 (1.0)*
Gender			
Male	51.2 (0.9)	54.2 (2.7)	50.8 (1.0)
Female	48.8 (0.9)	45.8 (2.7)	49.2 (1.0)
Insurance coverage status			
Currently insured	94.2 (0.5)	93.2 (1.4)	94.4 (0.6)
Currently uninsured	5.8 (0.5)	6.8 (1.4)	5.6 (0.6)
Race			
White alone	67.7 (0.9)	66.1 (2.7)	67.9 (1.0)
Black or African American alone	13.3 (0.6)	15.5 (1.9)	13.0 (0.7)
Other	19.0 (0.8)	18.3 (2.4)	19.1 (0.9)
Parent education			
Less than high school	9.6 (0.8)	12.6 (2.7)	9.2 (0.9)
High school or GED	20.6 (0.9)	24.4 (2.8)	20.0 (1.0)
Some college or technical school	22.0 (0.7)	21.1 (1.9)	22.1 (0.8)
College degree or higher	47.8 (0.9)	41.9 (2.6)	48.7 (1.0)
Developmental delay	6.3 (0.4)	13.4 (1.5)	5.4 (0.4)***
Intellectual disability	0.9 (0.1)	2.0 (0.6)	0.7 (0.1)**
Speech/language disorder	8.5 (0.5)	15.1 (1.7)	7.7 (0.5)***
Learning disability	7.1 (0.4)	12.4 (1.6)	6.3 (0.4)***
Autism/autism spectrum disorder	2.8 (0.3)	4.0 (0.7)	2.6 (0.3)*
ADD/ADHD	9.3 (0.4)	13.6 (1.6)	8.8 (0.4)**
Repeated grade	5.5 (0.5)	6.1 (1.0)	5.4 (0.5)
Special education/early intervention plan	9.5 (0.5)	12.7 (1.6)	9.1 (0.5)*

Note. p Values imply significant difference between preterm and full-term children in the sample: * $p < .05$, ** $p < .01$, *** $p < .001$. SE = standard error; ACE = adverse childhood experience; GED = general equivalency diploma; ADD/ADHD = attention deficit disorder/attention deficit hyperactivity disorder.

^aPercentages calculated for all children are based on 14,796 children with data of full-term/prematurity status.

Table 1 further shows the occurrence of neurodevelopmental outcome variables in all groups. Among all 6- to 11-year-olds in the 2016 NSCH data set, 6.3% had a developmental delay, 0.9% had an intellectual disability, 8.5% had a speech/language disorder, 7.1% had a learning disability, 2.8% had autism/ASD, 9.3% had ADD/ADHD, 5.5% repeated a grade, and 9.5% received special education/early intervention plan. Compared to children born full term, significantly more children born preterm experienced a developmental delay (13.4% vs. 5.4%, $p < .001$), an intellectual disability (2.0% vs. .7%, $p < .01$), a speech/language disorder (15.1% vs. 7.7%, $p < .001$), a learning disability (12.4% vs. 6.3%, $p < .001$), autism disorder/ASD (4.0% vs. 2.6%, $p < .05$), ADD/ADHD (13.6% vs. 8.8%, $p < .01$), and special education/early intervention (12.7% vs. 9.1%, $p < .05$).

Poverty

Table 2 shows the results of chi-square tests used to examine the relationship between neurodevelopmental outcomes, educational outcomes, and preterm birth and exposure to poverty. Odds ratios are reported in situations in which the results were statistically significant to quantify the magnitude of the difference between the groups and help with clinical interpretation.

Neurodevelopmental Outcomes As seen in Table 2, at 6–11 years of age, children born preterm and living in poverty had a 3.8 times greater risk of having a developmental delay; 19.3% of these children had a developmental delay, whereas 5.9% of the other children in the sample had a developmental delay. Children born preterm and living in poverty also had a 3.9 times greater risk of having an intellectual disability than did other children in the sample (3.1% vs. 0.8%). Compared to the other 6- to 11-year-olds in the study sample, 6- to 11-year-old children born preterm and living in poverty had a 2.7 times greater risk of speech/language disorders (19.4% vs. 8.2%), a 3.4 times greater risk of having a learning disability

(19.3% vs. 6.6%), a 2.4 times greater risk of ADD/ADHD (19.3% vs. 9.0%), and a 2.3 times greater risk of autism/ASD (6.0% vs. 2.7%).

Educational Outcomes Children born preterm and living in poverty had a 2.3 times greater risk of receiving services under a special education or early intervention plan compared to other children in the sample (19.1% vs. 9.2%). Children born preterm and experienced poverty were no more likely than other children to repeat a grade (9.1% vs. 5.4%).

Toxic Stress

Table 3 shows the results of chi-square tests used to examine the relationship between neurodevelopmental outcomes, educational outcomes, and preterm birth and exposure to toxic stress. Odds ratios are reported in situations in which the results were statistically significant to quantify the magnitude of the difference between the groups and help with clinical interpretation.

Neurodevelopmental Outcomes As seen in Table 3, at 6–11 years of age, children born preterm who had experienced toxic stress had a 3.6 times greater risk of having a developmental delay; 17.6% of the sample of children who were preterm and experienced toxic stress had a developmental delay, whereas 5.6% of the other children in the sample had a developmental delay. Children born preterm who had experienced toxic stress also had a 4.6 times greater risk of having an intellectual disability than other children in the sample (3.3% vs. 0.7%). Compared to the other 6- to 11-year-olds in the study sample, those born preterm and who had experienced toxic stress had a 2.5 times greater risk of speech/language disorders (17.8% vs. 8.0%), a 3.2 times greater risk of having a learning disability (18.0% vs. 6.3%), a 2.7 times greater risk of ADD/ADHD (20.4% vs. 8.6%), and a 2.5 times greater risk of autism/ASD (6.2% vs. 2.6%).

TABLE 2. Children Born Preterm and Living in Poverty

	Children born preterm and living in poverty	Other children	Odds ratio
Developmental delay***	19.3%	5.9%	3.8
Intellectual disability**	3.1%	0.8%	3.9
Speech/language disorder**	19.4%	8.2%	2.7
Learning disability***	19.3%	6.6%	3.4
Autism/autism spectrum disorder*	6.0%	2.7%	2.3
ADD/ADHD**	19.3%	9.0%	2.4
Repeated grade	9.1%	5.4%	
Special education/early intervention plan**	19.1%	9.2%	2.3

Note. Odds ratios are only shown when differences between groups are statistically significant. ADD/ADHD = attention-deficit disorder/attention-deficit/hyperactivity disorder.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

TABLE 3. Children Born Preterm With Toxic Stress

	Children born preterm and experienced toxic stress	Other children	Odds ratio
Developmental delay***	17.6%	5.6%	3.6
Intellectual disability***	3.3%	0.7%	4.6
Speech/language disorder***	17.8%	8.0%	2.5
Learning disability***	18.0%	6.3%	3.2
Autism/autism spectrum disorder***	6.2%	2.6%	2.5
ADD/ADHD***	20.4%	8.6%	2.7
Repeated grade	7.7%	5.4%	
Special education/early intervention plan**	15.3%	9.2%	1.8

Note. Odds ratios are only shown when differences between groups are statistically significant. ADD/ADHD = attention-deficit disorder/attention-deficit/hyperactivity disorder.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

Educational Outcomes Children born preterm who had experienced toxic stress had a 1.8 times greater risk of receiving services under a special education or early intervention plan compared to other children in the sample (15.3% vs. 9.2%). However, children born preterm who had experienced toxic stress were no more likely than other children to repeat a grade (7.7% vs. 5.4%).

Logistic Models

Tables 4 and 5 show the results of logistic regression models used to evaluate the effects of prematurity, toxic stress, and poverty on neurodevelopmental and educational outcomes while adjusting for a number of covariates. When controlling for toxic stress, poverty, and demographic factors, including poverty status, gender of the child, insurance status, child

TABLE 4. Logistic Regression: Neurodevelopmental Outcomes by Prematurity, Toxic Stress, and Covariates

	Developmental delay OR [95% CI]	Intellectual disability OR [95% CI]	Speech/language disorder OR [95% CI]	Learning disability OR [95% CI]	ADD/ADHD OR [95% CI]	Autism/ASD OR [95% CI]
Premature	2.72*** [2.06–3.60]	2.54** [1.29–5.00]	2.06*** [1.55–2.74]	1.89*** [1.39–2.58]	1.55** [1.19–2.03]	1.48 [0.97–2.26]
Toxic stress	1.97*** [1.52–2.54]	2.34** [1.25–4.39]	1.30* [1.00–1.68]	2.06*** [1.57–2.72]	2.60*** [2.10–3.22]	2.05*** [1.37–3.06]
Poverty (0–99% FPL)	1.42 [0.97–2.10]	1.70 [0.62–4.68]	1.20 [0.87–1.66]	1.64* [1.12–2.40]	1.14 [0.86–1.50]	1.48 [0.79–2.77]
Male	2.03*** [1.56–2.63]	3.68*** [2.00–6.80]	2.07*** [1.61–2.66]	1.74*** [1.32–2.31]	2.50*** [2.03–3.08]	5.36*** [3.54–8.11]
Uninsured	0.54 [0.29–1.01]	0.18** [0.05–0.63]	0.24*** [0.12–0.48]	0.52* [0.29–0.92]	0.38** [0.22–0.67]	0.40* [0.16–0.96]
Race (ref: White)						
Black	1.42 [0.97–2.07]	1.02 [0.34–3.01]	1.19 [0.83–1.70]	1.31 [0.89–1.95]	1.13 [0.81–1.58]	0.99 [0.55–1.78]
Other	0.95 [0.70–1.30]	0.72 [0.39–1.34]	0.74 [0.55–1.01]	0.61 [0.43–0.86]	0.74 [0.56–0.98]	1.47 [0.90–2.39]
Education (ref: College or higher)						
Less than high school	0.45* [0.23–0.88]	0.17 [0.02–1.73]	0.74 [0.41–1.35]	1.15 [0.65–2.03]	0.71 [0.42–1.20]	0.66 [0.26–1.69]
High school or GED	0.64 [0.45–0.91]	0.59 [0.24–1.41]	0.98 [0.65–1.47]	1.11 [0.76–1.60]	0.81 [0.61–1.08]	0.77 [0.42–1.38]
Some college or technical school	0.95 [0.69–1.30]	0.70 [0.28–1.72]	0.89 [0.67–1.19]	1.11 [0.80–1.54]	1.02 [0.80–1.29]	0.79 [0.47–1.32]

Note. Poverty (ref: 100% FPL or more); Race (ref: White); Parent education (ref: College degree or higher). OR = odds ratio; CI = confidence interval; ADD/ADHD = attention-deficit disorder/attention-deficit/hyperactivity disorder; ASD = autism spectrum disorder; FPL = federal poverty level; GED = general equivalency diploma.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

TABLE 5. Logistic Regression: Educational Outcomes by Prematurity, Toxic Stress, Poverty, and Covariates

	Repeated grade <i>OR</i> [95% CI]	Special education/early intervention plan <i>OR</i> [95% CI]
Premature	1.01 [0.67–1.54]	1.32 [0.98–1.76]
Toxic stress	1.37 [0.94–2.00]	1.39** [1.09–1.77]
Poverty (0–99% FPL)	1.85** [1.18–2.90]	1.53* [1.10–2.13]
Gender: Male	1.32 [0.89–1.95]	2.04*** [1.59–2.62]
Uninsured	1.29 [0.67–2.52]	0.43** [0.25–0.73]
Race (ref: White)	*	***
Black	0.97 [0.58–1.61]	1.60 [1.15–2.22]
Other	0.58 [0.37–0.91]	0.70 [0.51–0.95]
Education (ref: College degree or higher)	*	
Less than high school	2.21 [1.20–4.08]	0.68 [0.37–1.24]
High school or GED	2.18 [1.26–3.77]	0.91 [0.62–1.35]
Some college or technical school	1.15 [0.74–1.79]	1.10 [0.83–1.45]

Note. Poverty (ref: 100% FPL or more); Insurance coverage status (ref: Insured); Race (ref: White); Parent education (ref: College degree or higher). CI = confidence interval; FPL = federal poverty level; GED = general equivalency diploma.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

race, and parental education, children born preterm had increased odds of having a developmental delay ($OR = 2.72$, 95% CI [2.06, 3.60]), an intellectual disability ($OR = 2.54$, 95% CI [1.29, 5.00]), a speech/language disorder ($OR = 2.06$, 95% CI [1.55, 2.74]), a learning disability ($OR = 1.89$, 95% CI [1.39, 2.58]), and ADD/ADHD ($OR = 1.55$, 95% CI [1.19, 2.03]). Prematurity alone was not significantly related to autism, repeating a grade, and or receiving services under a special education or early intervention plan.

Controlling for prematurity, poverty, gender, insurance status, race, and parental education, being a child who experienced toxic stress increased the odds of having a developmental delay ($OR = 1.97$, 95% CI [1.52, 2.54]), an intellectual disability ($OR = 2.34$, 95% CI [1.25, 4.39]), a speech/language disorder ($OR = 1.30$, 95% CI [1.00, 1.68]), a learning disability ($OR = 2.06$, 95% CI [1.57, 2.72]), ADD/ADHD ($OR = 2.60$, 95% CI [2.10, 3.22]), autism/ASD ($OR = 2.05$, 95% CI [1.37, 3.06]), and receiving services through special education or early intervention plan ($OR = 1.39$, 95% CI [1.09, 1.77]). Undergoing toxic stress alone did not increase the likelihood of children in this sample to repeat a grade.

Controlling for prematurity, toxic stress, gender, insurance status, race, and parental education, being a child who experienced poverty increased the odds of a learning disability ($OR = 1.64$, 95% CI [1.12, 2.40]), repeating a grade ($OR = 1.85$, 95% CI [1.18, 2.90]), and receiving services through a special education or early intervention plan ($OR = 1.53$, 95% CI [1.10, 2.13]). Poverty was not significantly associated with experiencing a developmental delay, intellectual disability, speech or language program, autism, or ADHD.

Covariates Male children living in poverty were more likely to have a learning disability, to repeat a grade, and to receive services through special education or early intervention plan. Male children were also more likely to have a developmental delay, an intellectual disability, a speech/language disorder, a learning disability, ADD/ADHD, and autism/ASD and to receive services under a special education or early intervention plan. Being currently insured was positively associated with having an intellectual disability, speech/language disorder, learning disability, ADD/ADHD, and autism/ASD and receiving services under a special education or early intervention plan. Race was a significant predictor for having a learning disability, repeating a grade, and receiving special education/early intervention services. Children who identified as White were less likely than those identifying as Black to receive special education/early intervention services. Children who identified as White were more likely than children identifying as other (not identifying as Black or as White) to have a learning disability, to repeat a grade, and to receive special education/early intervention services. Children whose parents had a college degree or higher were more likely than children whose parents had less than a high school or high school/GED to have a developmental delay. Children with parents who had a college degree or higher were less likely than children with parents with less than a high school degree or high school/GED to repeat a grade.

Interactions To assess whether the effect of toxic stress differed for children born preterm versus full-term children, interaction terms were evaluated. Among 6- to 11-year-olds,

for ADHD and autism, significant interaction terms existed between prematurity and toxic stress ($p = .006$ and $p = .044$, respectively). Logistic regression models stratified by prematurity status showed that children born full term who experienced toxic stress were 2.4 times more likely to have ADD/ADHD than full-term children who did not experience toxic stress, and the influence of toxic stress was even greater for children born preterm. Specifically, children born preterm experiencing toxic stress were 4.6 times more likely to have ADD/ADHD than children born preterm with no toxic stress experiences. A similar pattern emerged for autism; children born full term who experienced toxic stress were 1.9 times more likely to have autism/ASD than full-term children without toxic stress experiences, and children born preterm who also experienced toxic stress were 3.6 times more likely to have autism/ASD than children born preterm with no toxic stress experiences.

DISCUSSION

It seems unlikely that the rate of preterm birth will be significantly reduced in the coming decade; therefore, healthcare providers, public policymakers, and education professionals need to understand the ramifications of preterm birth coupled with poverty and toxic stress. It has been recommended that research shifts from narrow outcome studies to multifaceted theoretical frameworks that recognize the effects of a child's environment and psychosocial situation on overall health (McCormick & Litt, 2017). Analysis of the 2016 NSCH data supports the EBD model of poverty and preterm birth, establishing the risk conferred to school-aged children born preterm who experience toxic stress and poverty.

In the current sample, 6- to 11-year-old children born preterm experienced significantly higher rates of all neurodevelopmental outcomes studied: developmental delay, speech/language disorder and learning disability, intellectual disability, ADD/ADHD, and special education/early intervention plan. These conditions are complicated by exposure to toxic stress and poverty, such that children born preterm who also experienced toxic stress and poverty suffered a disproportionately higher incidence of all neurodevelopmental and educational outcome variables except grade repetition, with odds ratios ranging from 4.6 for intellectual disability to 2.5 for autism/ASD. The current analysis suggests that White children are less likely than Black children to receive special education/early intervention services. This finding may reflect the disproportion of preterm birth experienced by Black women in the United States. Black women are 60% more likely to deliver preterm, a risk that has been linked to social determinants of health such as poverty, access to adequate healthcare, and nutrition (Mohamed, Thota, Browne, Diamond, & AHendy, 2014).

Interaction effects were seen for toxic stress and preterm birth, particularly related to ADD/ADHD and autism/ASD. These two conditions account for significant societal and

educational costs in the United States. Children and adults with attention-deficit/hyperactivity disorder have estimated yearly costs of ADD/ADHD (in 2010 dollars) ranging from \$9 to \$19 billion for educational support of children 5–18 years of age, \$4–\$14 billion in healthcare costs, and an estimated \$530 billion in juvenile detention and arrest costs for those whose impulsivity leads to involvement with the legal system (<http://www.chadd.org/Understanding-ADHD/About-ADHD/Data-and-Statistics/Cost-of-ADHD.aspx>). Costs for autism/ASD in the United States is projected to reach \$460 billion in 2025 assuming a prevalence of 1.1%, with the largest portion of costs being for education and parental lost productivity (Leigh & Du, 2015).

It is important to note that prematurity in the NSCH data is defined as birth before 37 weeks of gestation. The children in this study were born between 2005 and 2010, when the preterm birth rate ranged from 12.7% in 2005 to 9.98% in 2010. During those years, the vast majority of children born preterm were those born at 32–36 weeks of gestation, with just roughly 1.5%–2.1% born less than 32 weeks of gestation. As a nationally representative sample, it is likely, but not assured, that the children in this sample born preterm followed the pattern of preterm birth in the United States for those birth years.

Children born preterm, at all gestational ages, are at increased risk for impaired neurodevelopmental and educationally relevant outcomes during their elementary school years. Long-term outcome research with children born preterm links academic performance in childhood with wealth in adulthood (Basten et al., 2015) regardless of gestational age. Conventional recommendations include referral to early intervention services for children deemed to be at risk for developmental delays. However, most of the children born preterm—those born gestational ages between 32 and 36 weeks—do not typically meet state guidelines for automatic referral to early intervention services and neonatal follow-up and, in some settings, may not spend any time in the neonatal intensive care unit. Dempsey et al. (2015) recommended that all children born preterm receive evaluation at the time of school entry and periodically identify potential weaknesses that should be targeted for intervention before difficulties affect school performance. Health policy and public programs that support the family, encourage caregiver-child interactions, and address community-level sources of toxic stress are critical to addressing these risks (Garner et al., 2012; Johnson & Marlow, 2017).

The results reported here compel healthcare providers to address poverty and toxic stress in elementary school children, particularly those born preterm at all gestational ages, who appear to have a tripartite vulnerability—immaturity of body systems due to gestational age at birth, anatomic and functional brain changes related to poverty (Brumberg & Shah, 2015), and the detrimental physiological effects of toxic stress (Shonkoff et al., 2012). Our intent is not to support a fatalistic

view of the effects of poverty and ACEs on overall health; rather, we encourage programmatic interventions and policies that have the potential to address poverty, promote child well-being, and improve adult health (Wise, 2016). Most children born preterm are living independent and self-supportive lives and have self-perceived quality of life similar to term peers (Johnson & Marlow, 2017).

Limitations

Limitations of secondary data analysis include the inability to clarify participant responses and the need to select from existing variables. Causal relationship should not be inferred from cross-sectional data, such as the 2016 NSCH. The parent-reported prevalence of premature birth in this sample is consistent with reported *National Vital Statistics Reports* for those birth years; however, the prevalence cannot be validated with patient health records. The inability to describe the gestational age of the children and the severity of specific health conditions limits the ability to assess the degree to which those conditions affected study participants.

CONCLUSION

The current analysis used the EBD model of poverty and preterm birth to highlight the implications of poverty and toxic stress to the elementary school-aged child born preterm. Elementary education is a critical period for skill development and mastery, which may leave children born preterm who experienced toxic stress and poverty at risk and vulnerable to adverse neurodevelopmental outcomes. Healthcare providers need to take an active role in breaking the cycle of poverty, toxic stress, and preterm birth. To do this, recognition and intervention by primary care providers is essential.

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The Villanova University Institutional Review Board evaluated the research plan and deemed the study not subject to federal regulations under HHS 45 CFR 46.

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